

Analysis

UChicago Political Science Math Camp

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Analysis of Functions

Very rarely do we just want to “know” a function that we are working with. We might ask the following (why?):

- ▶ *What does a function do near some point?*
- ▶ *What does a function do as some input reaches infinity?*
- ▶ *What are the highest/lowest values a function might output?*

Some of these are easy. Others are fun!

What is a Limit?

Con conversationally, the **limit** represents the value a function approaches given an input behavior:

$$\lim_{x \rightarrow b} f(x)$$

represents the value that $f(x)$ approaches as x approaches b .

Some Limits Are Trivial

Focusing on finite limits, we can evaluate:



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$$\lim_{x \rightarrow a} \frac{a}{x}$$

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- ▶ *Hole discontinuities:*

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

Convergence & Divergence

A function **converges** if it gets very close to a constant value. Formally,

A function has a limit $\lim_{x \rightarrow p} f(x) = L$ iff for all $\epsilon > 0$, there exists some $\delta > 0$ such that $0 < |x - p| < \delta \implies |f(x) - L| < \epsilon$.

If there is no a for which this definition holds, then the sequence **diverges**.

What does this mean conversationally?



Some More Examples

$$\lim_{x \rightarrow \infty} x^2$$

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$$\lim_{x \rightarrow \infty} x^2$$

$$\lim_{x \rightarrow \infty} \sqrt{x}$$

There's Always Cauchy

If the limit itself is unknown, Cauchy Convergence provides an alternative definition:

A function f converges iff for all $\epsilon > 0$, there exists some δ such that $m, n > \delta \implies |f(m) - f(n)| < \epsilon$.

If the difference in subsequent values approaches zero, then the sequence/function must approach some constant.

An Illustrative Example: Redistribution

Boix (2003): Democracy is, in essence, a redistributive activity.

Suppose a state has fixed total wealth of 1. There is a poor class with wealth w_L , and an elite class with wealth w_H .

If redistribution from the wealthy to the poor happens repeatedly and fairly, what will ultimately happen with w_L and w_H ?

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Logically, what can this suggest about persistent inequality?

Existence of Limits

Let $\lim_{x \rightarrow a+} f(x)$ represent the right-side limit of $f(x)$ at a , and let $\lim_{x \rightarrow a-} f(x)$ represent the left-side limit.

$$\lim_{x \rightarrow a} f(x) \text{ exists iff } \lim_{x \rightarrow a+} f(x) = \lim_{x \rightarrow a-} f(x).$$

This tends to affect two types of functional features:

1. Piecewise
2. Asymmetric vertical asymptotes

End Behavior

We are often interested in the “**end behavior**” of a function; i.e. its composition approaching infinity:

$$\text{End behavior of } f(x) \equiv \lim_{x \rightarrow \infty} f(x)$$

This usually results in $\lim_{x \rightarrow \infty} f(x) \in \{-\infty, a \in \mathbb{R}, \infty\}$

Continuity

Fundamentally, **continuity** is thought of as meaning “one continuous line.” But, is a function really a line?

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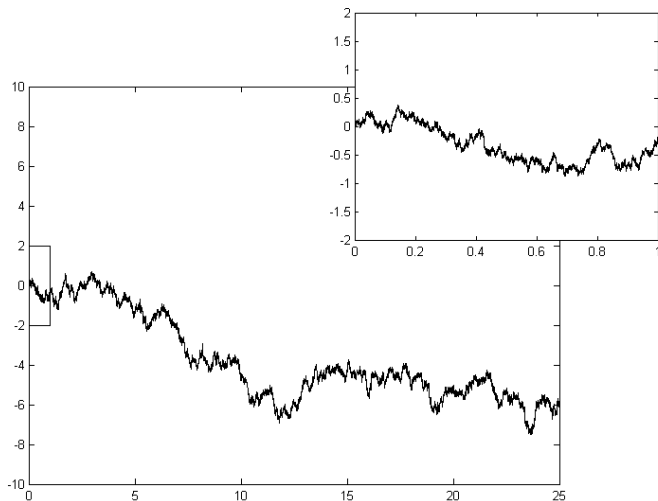
A function is continuous at point c if $\lim_{x \rightarrow c} f(x) = f(c)$.

Differentiability

A function is **differentiable** over $[a, b]$ if for every $c \in [a, b]$, there exists a rate of change $m \in \mathbb{R}$.

(We will enhance this definition later on using derivatives.)

A Cool Math Fact: Brownian Motion



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Extrema

The **maxima** (**minima**) of a set is the element(s) of that set for which no other element is **greater** (**lesser**).

(This requires a well-ordering principle, which is a property of $\mathbb{R}$ and its subsets.)

Maxima & Minima

For most well-defined sets, finding maximal and minimal elements is straightforward

We will see in Optimization that evaluating these for function co-domains is conceptually straightforward (even if technically challenging)

We use the following notation in such cases (sub *max* = *min*):

$\max\{X\}$ = Maximal element in set X

$\max f(x)$ = Maximal output of $f(x)$

$\arg \max_x f(x)$ = Value of x which maximizes f

Can We Break This?

Let us define the following bijective function such that every real number is the output for a unique input:

$$f : \mathbb{R}_+ \rightarrow \mathbb{R}_+ \qquad f(x) = \frac{1}{x}$$

We then evaluate:

$$\max f(x)$$

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$$\min f(x)$$

We Broke It!

We can (and very frequently do) create well-defined sets which are **missing a well-defined extrema**

What aspect of our function's domain (along with its bijectivity) suggests this dilemma?

How can we define a set's domain without defining the most extreme numbers it contains?

Some Numbers Are Big And Others Are Small

To solve this problem, we start with a rather dumb idea:

A **bound** is an element on either side of a set.

- ▶ A **lower bound** is less than or equal to than every element
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Bounds are not unique. Define $X = \{x \in \mathbb{R} : 1 < x < 2\}$. We can call 0.89 a **lower bound**, and π an **upper bound**.

An Interesting Idea

We may not know every number the set contains. But, we do know what numbers the set does **not** contain...

Define the set of numbers which are not contained by our function $f(x) = \frac{1}{x}$ over the domain $x \in \mathbb{R}_+$.

Two Important Bounds

- ▶ The **greatest lower bound** of a set A is a number b_0 such that b_0 is a lower bound of A and for any other lower bound b , then $b \leq b_0$.
- ▶ The **least upper bound** of a set A is a number b_0 such that b_0 is an upper bound of A and for any other upper bound b , then $b \geq b_0$.

We denote the **greatest lower bound** as the **infimum**, denoted $\inf(A)$, and the **least upper bound** as the **supremum**, denoted $\sup(A)$.

The Smallest Positive Number?

We have seen that $\min\{\frac{1}{x} : x \in \mathbb{R}_+\}$ is not well defined: we cannot create a smallest positive number.

But, we can identify a greatest lower bound of this function.

$$\implies \inf\left(\left\{\frac{1}{x} : x \in \mathbb{R}_+\right\}\right) = 0$$

There is no smallest positive number, but there is a largest non-positive number!

Intermediate Value Theorem

A simple-yet-powerful analytic tool:

If f is continuous over $[a, b]$, then for every $y \in [f(a), f(b)]$, there exists some x such that $f(x) = y$.

Think: If you take University from 59th St to 58th St, you HAVE to pass 5828 University (Pick Hall!)

IVT Lends Existence

We often use IVT to prove that some threshold or boundary exists (even if we don't know what it is):

"I know that (A) is true at ($x=a$) and (B) is true at ($x=b$). Then, there is some value of x where (A) switches to (B)."

Suppose I can offer cash to a previously party-A voter to convince them to vote for party-B. For some amount of cash, I observe them vote for party B. *What does that tell me about their beliefs?*